Actual twist relation yield and main-source recovery

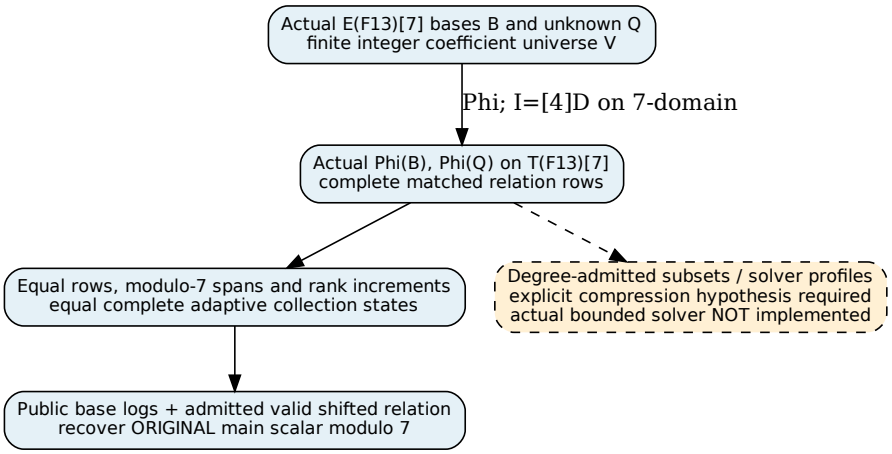
2026-10-30 PR2381 20 named theorems /24 audited declarations

Actual objects. Curve field $\mathbf{F}_{13}$: $E : y^2 = x^3 + x + 4$, $T : y^2 = x^3 + 4x + 6$,the actual monic $d = 2$ twist. $P = (0, 2) \mapsto U = (8, 11)$,order7. The prior checked coordinate maps are Φ and $I = [4]D$; $I\Phi Q = Q$ on $E[7]$. These are Mathlib nonsingular points and coordinate functions,not target-log lookup tables.

Complete matched relation theorem. For any finite integer coefficient universe V,bases $B_i \in E[7]$ and unknown $Q \in E[7]$,

$$\sum_i v_i \Phi(B_i) = \Phi(Q) \quad \Longleftrightarrow \quad \sum_i v_i B_i = Q.$$

Complete matched row sets/cardinalities coincide. Reduce coefficients modulo7: the relation-field $\mathbf{F}_7$ spans and incremental ranks $\dim(L + \langle R \rangle) - \dim L$ agree for every coefficient state$L$. A supplied adaptive query rule must stay in$E[7]$;complete matched collection steps,their finite iterates and rank progress are then equal.



Domain control. Global injectivity is false:actual $K = (10, 0) \neq 0$, $\Phi K = 0$,and$7K \neq 0$,all kernel checked. The seven-torsion hypotheses are material. Aggregate inverse recovery does not assert individual summands vanish.

From an admitted relation to the original scalar

For explicit *public setup logs* $A_i = \ell_i U$ and an accepted actual relation

$$\sum_i v_i A_i = \Phi(Q) + sU, \qquad Q \in E[7],$$

Lean proves for signed integers

$$Q = \Big(\sum_i v_i \ell_i - s\Big)P = \Big(\big(\sum_i v_i \ell_i - s\big) \bmod 7\Big)P.$$

Unknown$Q$ is an actual arbitrary source point in the stated domain. Setup logs must be legitimately derived;no target scalar is supplied to a solver. This gives an actual main-DLP bridge for admitted relations,not a measurement of relation finding or free setup/linear algebra.

Degree qualification is a separate obligation. The new module allows abstract numerical profiles$d_E(v), d_T(v)$ and caps. An explicit compression hypothesis ensures source-admitted rows meet the target cap;only then is source qualified span contained in target qualified span. No claim identifies those profiles with the native solver,degree of regularity or runtime. Complete row equality alone supplies no admitted-subset or cost equality. A higher qualified yield/lower relevant degree still needs actual solver evidence and full references.

What was run. The unmodified scoped repository gate passes24declarations, 20named plus4generated,zero failing/extra trusted/native axioms. Only `propext,Classical.choice,Quot.sound`. Lean4.34.1/Mathlibd13f23b7. Six builds:three failed,three successful progressive stages. Failed sources/logs remain;same statements corrected for action/coercion and variable-scalar sum lemmas. Explicit finite-set induction closes the final signed recovery proof. Historical imported receipts are not a fresh global audit. All18old polynomial certificates remain unchanged;no field IDC is substituted for group proofs.

What has not run. The separate all16F13arm bounded-solver study (original/reduced,caps0–8,all6targets,blocked setups,strongest-source/matched/ direct-enumeration controls) is *unimplemented and unexecuted*. It is a validated draft card,not a ready/funded/Cairn task. No new native census, scalar recoveries,degree/yield/speed measurement or external checker is claimed. This proof does not close generic PairRank independence,native iid stopping-time semantics,absorption/integrability,broader geometry or calibrated complete work.

Open gates. Latest root5140-case suite remains incomplete600s/124; 1921study registrations,naming,required exact-head CI/review/integration and Conductor127/no lease remain held. No registry/catalog/benchmark/leaderboard or old frozen record changed. Keep PR open/non-draft and auto-mergeoff. Proof caps 600sec,iteration03:08–04:08UTC,two localworkers,no paidcompute/broaderassetupload.

Source and raw restoration links are in the adjacent README. Literature: prior CM8/dual,PhaseRecovery and conditional Relation-Cost modules;lead-atlas isogeny-transported, decomposition and degree leads. No invention or universal no-advantage,hardness,GHS,practical-speedup or invalid-curve claim.